HAMJSCC: Hierarchical Adaptive Multi-Path Visual Transmission and Reconstruction for Dynamic IoV Scenarios

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Abstract. Wireless image transmission in dynamic Internet of Vehicles (IoV) environments is challenged by fluctuating environmental conditions. Existing schemes struggle to jointly balance reconstruction fidelity, perceptual quality, and efficiency. Although generative schemes can improve perceptual quality, they usually incur high inference latency. To address these issues, this study proposes a hierarchical adaptive multi-path (HAMJSCC) framework for visual transmission systems in IoV. Within the proposed framework, signal-to-noise ratio (SNR) and channel bandwidth ratio (CBR) are introduced for environment-adaptive feature transmission, where dynamic truncation rather than masking is proposed to achieve truly bandwidth-constrained transmission. The decoding branch is dynamically selected according to environmental conditions. A Transformer-based branch is adopted under favorable conditions, while an edge-assisted generative branch is employed under harsh conditions to improve visual perceptual quality and reduce device-side inference latency. Experimental results demonstrate that HAMJSCC achieves comparable PSNR performance to existing schemes while reducing LPIPS by 15%–25%. Compared with diffusion-based schemes, HAMJSCC further reduces inference latency by approximately 75%.

Keywords: Visual Transmission System · Image Reconstruction · Environment Adaptation · Edge-assisted Generation.

1 Introduction

With the advancement of artificial intelligence and wireless communication technologies, wireless image transmission [2] has become a key support for intelligent visual transmission systems [5, 12, 14] and is widely used in scenarios such as the Internet of Vehicles (IoV) [22, 9, 10] and edge intelligence [8, 3]. Traditional schemes [11] adopt separate source and channel coding structures [13], which are prone to error propagation and quality degradation in complex wireless environments. Joint source–channel coding (JSCC) [16] addresses this problem through joint optimization, demonstrating stronger robustness under low signal-to-noise ratio (SNR) and limited channel bandwidth ratio (CBR), and has attracted attention in recent years.

In terms of environment adaptation, existing schemes fall into two categories: SNR-adaptive [15, 18] and CBR-adaptive [1, 17]. The former leverages real-time SNR to dynamically modulate features, improving robustness under varying SNR. The latter mostly relies on masking mechanisms, where inactive parts still occupy resources, failing to achieve true limited-bandwidth transmission. Therefore, how to jointly adjust the transmission strategy based on SNR and available bandwidth in dynamic environments remains a key issue.

At the receiver side, existing schemes typically employ fixed decoding architectures for image reconstruction, such as CNNs [2, 15], Transformers [1, 18, 17], and generative models. Although different architectures exhibit their own advantages in terms of reconstruction fidelity, perceptual quality, and computational complexity, most existing schemes still rely on static decoding strategies and cannot dynamically adjust the decoding process according to environmental conditions. As a result, single-path fixed decoding schemes struggle to satisfy diverse image reconstruction requirements in complex wireless environments.

Furthermore, from the perspective of system architecture and deployment, existing schemes typically perform image reconstruction and inference entirely on the device side. In particular, generative schemes introduce complex models, such as GANs [6] and diffusion models [7], to improve visual perceptual quality. However, these models usually incur high computational overhead and inference latency, leading to significant resource consumption on devices. Therefore, in dynamic scenarios such as IoV, existing schemes still struggle to balance perceptual quality and system efficiency.

To address the above issues, this study proposes a hierarchical adaptive multi-path wireless image transmission framework for visual transmission systems. The framework jointly exploits SNR and CBR for environment-adaptive modeling, and dynamically selects decoding branches according to the environmental state at the decoding stage. Through the above design, it achieves synergistic optimization among reconstruction fidelity, perceptual quality, and computational efficiency. The main contributions are as follows:

- A hierarchical adaptive multi-path wireless image transmission framework for visual transmission systems is proposed. It dynamically selects decoding branches based on environmental conditions: an edge-assisted generative decoding branch in harsh environments, and a Transformer branch under favorable conditions.
- A dual-condition environment-adaptive modulation/demodulation mechanism is designed to jointly model image semantic features with SNR and CBR. Through dynamic channel pruning at the encoder and conditional decoupling at the decoder, transmission efficiency and reconstruction quality are improved in complex environments.
- An edge-assisted generative refinement mechanism is proposed to build a collaborative decoding scheme of “edge-based fidelity recovery and device-based generative refinement”. By leveraging latent space transmission and downlink SNR conditioning, perceptual quality and robustness are enhanced under harsh conditions.

2 RELATED WORK

A comparison of representative schemes is provided in Table 1.

Table 1. Comparison of Related Studies

Ref.	Model	Decoder	SNR	CBR	Gen.	Architecture
[2]	DeepJSCC	CNN	×	×	×	Device-Only
[15]	ADJSCC	CNN	Aware	×	×	Device-Only
[18]	WITT	Transformer	Aware	×	×	Device-Only
[1]	DeepJSCC++	Transformer	Aware	Mask	×	Device-Only
[17]	SwinJSCC	Transformer	Aware	Mask	×	Device-Only
[19]	DeepJSCC-adv	CNN	Aware	×	✓	Device-Only
[4]	InverseJSCC	CNN	Aware	×	✓	Device-Only
[21]	DeepJSCC-Diff	CNN	Aware	×	✓	Device-Only
[20]	DiffJSCC	Transformer	Aware	×	✓	Device-Only
Ours	HAMJSCC	**Multi-Path**	Aware	**Trunc.**	✓	**Hierarchical**

2.1 Environment Adaptation

Existing schemes have evolved from fixed to adaptive approaches. Early DeepJSCC [2] did not explicitly consider dynamic channel conditions. ADJSCC [15] and WITT [18] introduced SNR-adaptive feature transmission. DeepJSCC-l++ [1] and SwinJSCC [17] further incorporated CBR to achieve dual adaptation of SNR and CBR, but still rely on masking mechanisms and fail to realize truly limited-bandwidth transmission.

2.2 Decoding Architecture

Early CNN-based schemes (DeepJSCC [2], ADJSCC [15]) suffer from insufficient detail recovery. Subsequent Transformer-based schemes (WITT [18], DeepJSCC-l++ [1], SwinJSCC [17]) leverage self-attention to enhance global modeling and improve reconstruction quality. Some works adopt GANs or diffusion models for generative decoding, but none of them can adaptively select decoding branches according to the environment.

2.3 Generative Reconstruction

DeepJSCC-adv [19] and InverseJSCC [4] adopt GANs to enhance texture details. DeepJSCC-Diff [21] and DiffJSCC [20] introduce diffusion models to achieve more stable perceptual reconstruction under low SNR and low CBR conditions. However, these generative schemes perform inference solely on the device side without edge assistance, leading to excessive computational load and high inference latency.

3 SYSTEM MODEL

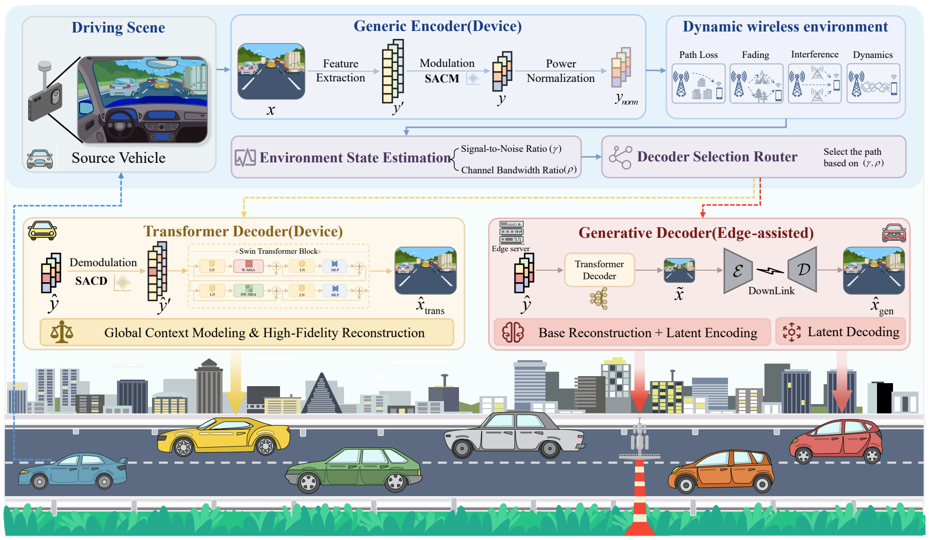

Fig. 1. Overview of the proposed HAMJSCC framework.

3.1 Overall Architecture of HAMJSCC

As shown in Fig. 1, the transmitter-side generic encoder consists of a feature extraction module and the proposed Semantic Adaptive Condition Modulator (SACM). The feature extraction module extracts multi-scale semantic features y' from the input image x, and SACM further incorporates environment state information to adaptively modulate y', generating a transmission symbol sequence y, which is then power-normalized to obtain the transmit signal y_{norm}.

The receiver obtains the noisy received signal $\hat{y}$. The Semantic Adaptive Condition Demodulator (SACD) demodulates $\hat{y}$ based on the environment state information and recovers the semantic features $\hat{y}'$. Then, the decoding path is selected according to SNR γ and CBR ρ, with the rule defined as follows:

$$\text{route}(\gamma, \rho) = \begin{cases} \text{Generative}, & \gamma \leq \frac{\gamma_{\max}}{2} \text{ and } \rho \leq \frac{\rho_{\max}}{2}, \\ \text{Transformer}, & \text{otherwise}, \end{cases} \tag{1}$$

where $\gamma_{\max} = 15$ dB and $\rho_{\max} = 1/16$ are set in this study, the edge-assisted generative branch is enabled when the environment is harsh to output $\hat{x}_{\text{gen}}$ with high fidelity and high perceptual quality; otherwise, the Transformer branch is adopted to output $\hat{x}_{\text{trans}}$ with high fidelity.

3.2 Semantic Adaptive Condition Modulator / Demodulator

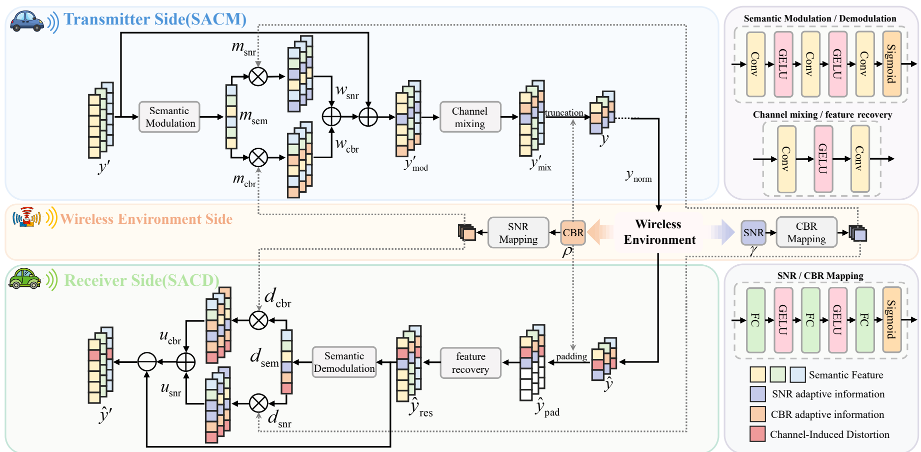

Fig. 2. SACM / D: Semantic Adaptive Condition Modulator / Demodulator

To enhance adaptability in complex wireless environments, this study proposes the Semantic Adaptive Condition Modulator (SACM) and Demodulator (SACD), with their structure shown in Fig. 2. At the transmitter, SACM deeply integrates semantic features with environmental state, dynamically adjusting feature strength and transmission dimension to achieve efficient limited-bandwidth transmission. At the receiver, SACD suppresses environmental disturbances and noise to achieve high-fidelity recovery of semantic information.

The modulator jointly models the semantic features y' extracted from the input image x with SNR γ and CBR ρ. Through an SNR mapping network $f_{\text{snr}}(\cdot)$, a CBR mapping network $f_{\text{cbr}}(\cdot)$, and a semantic modulation network $f_{\text{m}_{\text{sem}}}(\cdot)$, it generates the corresponding environmental modulation coefficients m_{snr}, m_{cbr} and the semantic modulation coefficient m_{sem}, respectively. This process can be expressed as:

$$m_{\text{snr}} = f_{\text{snr}}(\gamma), \quad m_{\text{cbr}} = f_{\text{cbr}}(\rho), \quad m_{\text{sem}} = f_{\text{m}_{\text{sem}}}(y'). \tag{2}$$

Then, the semantic modulation coefficient is element-wise multiplied with the SNR and CBR modulation coefficients, respectively, to obtain the corresponding conditional modulation features:

$$w_{\text{snr}} = m_{\text{sem}} \odot m_{\text{snr}}, \quad w_{\text{cbr}} = m_{\text{sem}} \odot m_{\text{cbr}}. \tag{3}$$

The input semantic features y' are jointly and adaptively enhanced via residual modulation, producing the modulated feature y'_{mod}:

$$y'_{\text{mod}} = y' + \kappa_{\text{snr}} w_{\text{snr}} + \kappa_{\text{cbr}} w_{\text{cbr}}, \tag{4}$$

where κ_{snr} and κ_{cbr} are learnable modulation intensity parameters that control the influence of different conditional information on the semantic features.

Furthermore, to enhance information interaction among different semantic channels, we introduce a channel mixing network that performs cross-channel fusion on the modulated features. Based on the available bandwidth, only the top-C_t most important channels are dynamically retained for transmission, forming the final transmission symbol sequence y:

$$y'_{\text{mix}} = f_{\text{mix}}(y'_{\text{mod}}), \quad y = y'_{\text{mix}}[:,:,:C_t], \tag{5}$$

where $f_{\text{mix}}(\cdot)$ is the channel mixing network, and C_t is the number of available transmission channels. This dynamic channel truncation mechanism enables the system to adaptively adjust the amount of transmit information according to the real-time bandwidth, achieving truly limited-bandwidth transmission.

After modulation, y is power-normalized to obtain the transmit signal y_{norm}. Once this signal enters the wireless environment, the received signal $\hat{y}$ suffers from semantic distortion and perturbations. SACD demodulates $\hat{y}$, with a structure symmetric to the modulator, jointly modeling SNR, CBR, and $\hat{y}$ to dynamically recover damaged semantics and suppress environmental disturbances.

At the receiver side, the received signal $\hat{y}$ is first adaptively padded to the original size of the transmission symbol sequence y. Then, cross-channel reconstruction and semantic completion are performed on the received signal $\hat{y}$ via a feature recovery network to obtain the initial recovered features:

$$\hat{y}_{\text{pad}} = f_{\text{pad}}(\hat{y}), \quad \hat{y}_{\text{res}} = f_{\text{res}}(\hat{y}_{\text{pad}}), \tag{6}$$

where $f_{\text{pad}}(\cdot)$ denotes the feature padding operation, and $f_{\text{res}}(\cdot)$ denotes the feature recovery network.

The demodulator reuses the SNR and CBR mapping coefficients obtained in the modulation stage. To further capture the semantic distribution in the recovered features, a semantic demodulation network is adopted, which produces the following coefficients:

$$d_{\text{snr}} = m_{\text{snr}}, \quad d_{\text{cbr}} = m_{\text{cbr}}, \quad d_{\text{sem}} = f_{\text{d}_{\text{sem}}}(\hat{y}_{\text{res}}), \tag{7}$$

where $f_{\text{d}_{\text{sem}}}(\cdot)$ is the semantic demodulation network at the receiver.

Then, the semantic demodulation coefficient d_{sem} is element-wise multiplied with the SNR and CBR demodulation coefficients, respectively, to obtain the conditional demodulation features:

$$u_{\text{snr}} = d_{\text{sem}} \odot d_{\text{snr}}, \quad u_{\text{cbr}} = d_{\text{sem}} \odot d_{\text{cbr}}. \tag{8}$$

Finally, the demodulator suppresses environmental disturbances in $\hat{y}_{res}$ via residual subtraction to obtain the demodulated semantic features $\hat{y}_{\text{dsem}}$:

$$\hat{y}_{\text{dsem}} = \hat{y}_{res} - \kappa'_{\text{snr}} u_{\text{snr}} - \kappa'_{\text{cbr}} u_{\text{cbr}}, \tag{9}$$

where κ'_{snr} and κ'_{cbr} are learnable demodulation intensity parameters that dynamically control the influence of different environmental conditions on semantic recovery.

3.3 Generative Decoding Branch

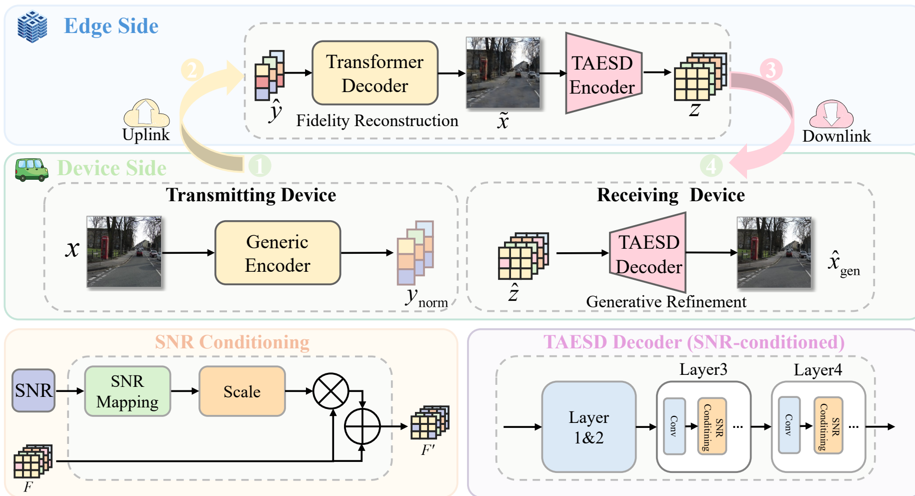

Fig. 3. Generative Decoding Branch

At the receiver, to address semantic distortion and detail degradation under harsh environments, we design a generative refinement branch, whose structure is shown in Fig. 3. Unlike traditional pixel-wise reconstruction schemes, this branch is built upon a pre-trained generative model, TinyVAE (TAESD), and leverages its latent space prior. Through a collaborative mechanism of "edge-based fidelity recovery and device-based generative refinement", it achieves consistent restoration of texture details, edge structures, and global semantics under complex degradation conditions.

Specifically, at the transmitting device, the input image x is first processed by the Generic encoder to obtain the normalized transmit signal y_{norm}. This signal is then transmitted to the edge side over the uplink wireless environment. At the edge side, the received signal $\hat{y}$ is fed into a Transformer-based decoder to reconstruct the basic image $\tilde{x}$. The overall process can be expressed as:

$$\underbrace{y_{\text{norm}} = \text{Encoder}(x),}_{\text{Device}} \qquad \hat{y} = \text{Uplink}(y_{\text{norm}}), \qquad \underbrace{\tilde{x} = \text{Decoder}(\hat{y}),}_{\text{Edge}} \tag{10}$$

where Uplink$(\cdot)$ models the general wireless environment, equivalent to a standard device-to-device link.

After obtaining the basic reconstruction result $\tilde{x}$, to reduce transmission load and improve the efficiency of subsequent generative recovery, the TAESD encoder is introduced to map the basic reconstructed image into the latent space, yielding the latent variable:

$$z = \mathcal{E}_{\text{edge}}(\tilde{x}), \tag{11}$$

where $\mathcal{E}_{\text{edge}}$ is a pre-trained and frozen lightweight VAE encoder, and z is the corresponding latent variable.

Then, the edge side sends the latent variable z to the receiving device over the downlink.

$$\hat{z} = h_{\text{d}}(z) + n_{\text{d}}, \tag{12}$$

where $h_{\text{d}}(\cdot)$ denotes the downlink transmission process from the edge to the receiving device, n_{d} is the additive noise, and ẑ is the latent variable received by the receiving device.

Finally, the receiving device performs generative reconstruction on the received latent variable $\hat{z}$ using the proposed SNR-conditioned TAESD decoder:

$$\hat{x}_{gen} = \mathcal{D}_{\text{device}}(\hat{z}, \gamma_{\text{d}}), \tag{13}$$

where $\mathcal{D}_{\text{device}}(\cdot)$ denotes the TAESD decoder that reconstructs $\hat{x}_{\text{gen}}$ from the received latent variable $\hat{z}$, conditioned on the downlink SNR γ_{d}.

To implement this conditioning mechanism, we further describe the internal design of the TAESD decoder. Latent transmission relies on a pre-trained VAE with a fixed compression ratio of 1/24, while high-level feature refinement is conditioned only on the downlink SNR γ_{d}. To realize this, γ_{d} is first mapped into a channel-wise conditioning vector c_{snr}:

$$c_{\text{snr}} = f_{\text{snr}}(\gamma_{\text{d}}). \tag{14}$$

To avoid excessive influence of conditional information on features, which may cause instability or information loss, we introduce a residual gating mechanism to regulate its strength. This improves stability while smoothly integrating the conditioning signal. The resulting channel-wise gating coefficients are given as follows:

$$\alpha = 1 + G \odot (2\beta(c_{\text{snr}} - 0.5)), \tag{15}$$

where G is a learnable gating parameter and β is a scaling coefficient. This module is embedded in the last two upsampling stages of the TAESD decoder.

Let F denote the original high-level feature map to be gated in the decoder (obtained from the previous stage). The gated feature map F' is then obtained via channel-wise rescaling:

$$F' = \alpha \odot F. \tag{16}$$

4 EXPERIMENTAL RESULTS

4.1 Experimental Setup

Three experiments are conducted to evaluate the proposed framework. The first analyzes the reconstruction performance of HAMJSCC-Base (single Transformer branch). The second evaluates the reconstruction quality and perceptual performance of HAMJSCC (Transformer and generative decoding) under harsh environments. The third, reported in Appendix A, evaluates the edge-assisted generative branch under dynamic uplink and downlink environments. Numerical results of all main figures are provided in Appendix B.

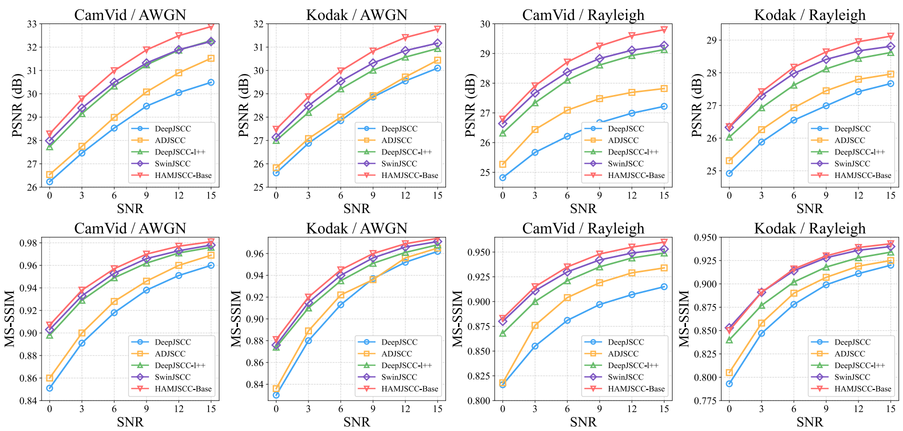

Fig. 4. Comparison of PSNR and MS-SSIM performance of different schemes under AWGN and Rayleigh environments. The CBR is 1/32 for the CamVid dataset and 1/24 for the Kodak dataset.

This study follows the common practice in the field of wireless image transmission, and all experiments are conducted for image reconstruction. For the IoV scenario, the CamVid dataset is used, with 367 images for training, 101 for validation, and 233 for testing, with center cropping to 896×640. For natural scenes, DIV2K (800 images) is used for training and validation, and Kodak (24 images) for testing. During training, all images are randomly cropped to 256×256. Complexity and latency metrics are all computed at this resolution.

The experiments are implemented with PyTorch 2.7.1 and trained on an NVIDIA L20 (48 GB). Training consists of two stages: first, HAMJSCC-Base is trained with a learning rate of 10^{-4}; then, the encoder and Transformer branch are frozen, and only the newly added modules of the generative decoder are optimized. The optimizer is Adam with a batch size of 8, and other hyperparameters remain unchanged. A collaborative simulation environment is built with an edge (NVIDIA L20) and a device (RTX 4060).

4.2 Evaluation of HAMJSCC-Base with Transformer-Only Decoder

As shown in Fig. 4, under the same CBR, the reconstruction quality of all schemes increases monotonically with SNR. On the CamVid dataset under the AWGN environment (CBR=1/32), HAMJSCC-Base achieves a PSNR improvement from 28.28 dB to 32.88 dB, which is 1.8–2.4 dB higher than DeepJSCC, with an MS-SSIM of up to 0.981. On the Kodak dataset under the AWGN environment (CBR=1/24), its PSNR increases from 27.48 dB to 31.77 dB, outperforming SwinJSCC by an average of 0.6–0.8 dB. Under the Rayleigh environment, the advantage remains stable. For example, on CamVid, it reaches 29.80 dB in PSNR and 0.960 in MS-SSIM, both better than the compared schemes.

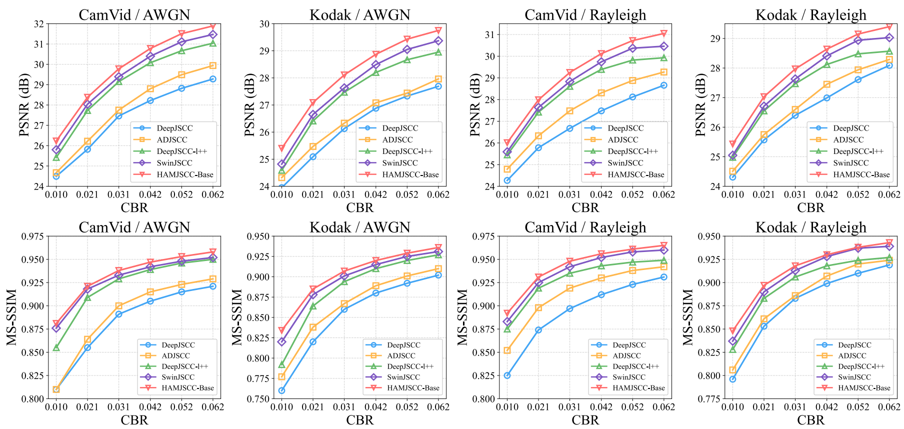

Fig. 5. Comparison of PSNR and MS-SSIM performance of different schemes on the CamVid and Kodak datasets (AWGN environment: SNR=3 dB; Rayleigh environment: SNR=9 dB).

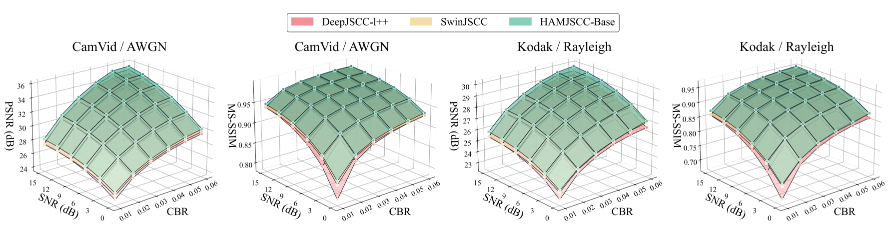

Fig. 6. PSNR and MS-SSIM performance of different schemes on CamVid under AWGN environment and on Kodak under Rayleigh environment.

As shown in Fig. 5, under the same SNR, the reconstruction performance of all schemes improves as CBR increases. On the CamVid dataset under the AWGN environment (SNR = 3 dB), the PSNR of HAMJSCC-Base increases from 26.23 dB to 31.89 dB, which is on average 1.5–2.0 dB higher than DeepJSCC, with the highest MS-SSIM reaching 0.958. On the Kodak dataset under the AWGN environment, its PSNR rises from 25.40 dB to 29.75 dB, stably outperforming SwinJSCC by about 0.3–0.6 dB. Under the Rayleigh environment (SNR = 9 dB), this improvement trend is further enhanced.

As shown in Fig. 6, under joint variations of CBR and SNR, HAMJSCC-Base consistently maintains higher reconstruction quality. For example, on CamVid AWGN, its PSNR reaches 28–35 dB and MS-SSIM reaches 0.988, outperforming compared schemes. On Kodak Rayleigh, it maintains a smoother performance surface in low-CBR and low-SNR regions, showing stronger degradation resistance. In summary, the proposed scheme achieves better global performance distribution and stability under complex environmental variations.

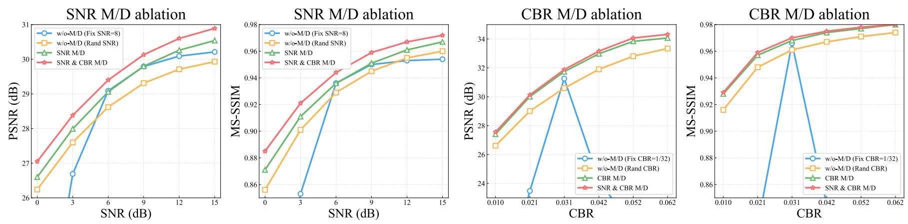

Fig. 7. Ablation study results of the proposed SACM/D on CamVid dataset under AWGN environment. The first two subfigures demonstrate the effectiveness of SNR-adaptive modulation/demodulation, and the last two subfigures demonstrate the effectiveness of CBR-adaptive modulation/demodulation.

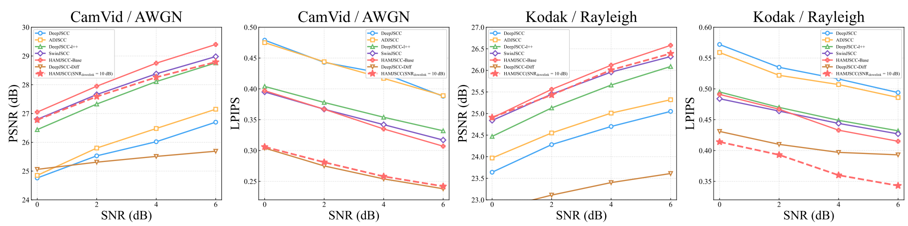

Fig. 8. Under harsh condition of CBR = 1/48, the PSNR and LPIPS performance of different schemes on CamVid under AWGN environment and on Kodak under Rayleigh environment.

As shown in Fig. 7, the dual-condition adaptive modulation/demodulation (SNR & CBR M/D) outperforms fixed, random, and single-condition adaptive (SNR-only or CBR-only) schemes in both PSNR and MS-SSIM. Specifically, on CamVid under AWGN environment, it achieves a PSNR of 30.89 dB and MS-SSIM of 0.972; on Kodak under Rayleigh environment, it achieves 34.32 dB and 0.980, respectively. Fixed or random configurations suffer from significant performance degradation under extreme conditions. These results demonstrate that the proposed SACM/D module, by jointly modeling SNR and CBR environmental information, significantly improves system robustness and reconstruction quality, enabling truly adaptive limited-bandwidth transmission.

4.3 Evaluation of HAMJSCC

As shown in Fig. 8, under a fixed CBR of 1/48, HAMJSCC achieves slightly lower PSNR than SwinJSCC but significantly better LPIPS (0.240 vs. 0.317). DeepJSCC-Diff obtains even lower LPIPS (0.238) but at the cost of a much lower PSNR of only 25.69 dB. On Kodak Rayleigh, HAMJSCC achieves the highest PSNR (26.39 dB) and the lowest LPIPS (0.343). In summary, HAMJSCC trades off a marginal loss in pixel-wise fidelity for substantially improved perceptual quality, while avoiding the severe PSNR collapse seen in pure generative schemes, thus achieving a favorable balance between perceptual quality and fidelity.

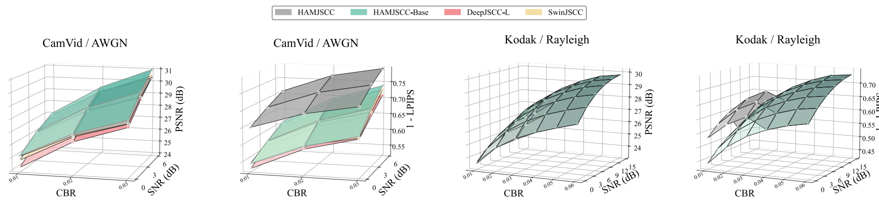

Fig. 9. Left: Performance of different schemes on CamVid AWGN under harsh conditions; right: global comparison of HAMJSCC and HAMJSCC-Base on Kodak Rayleigh.

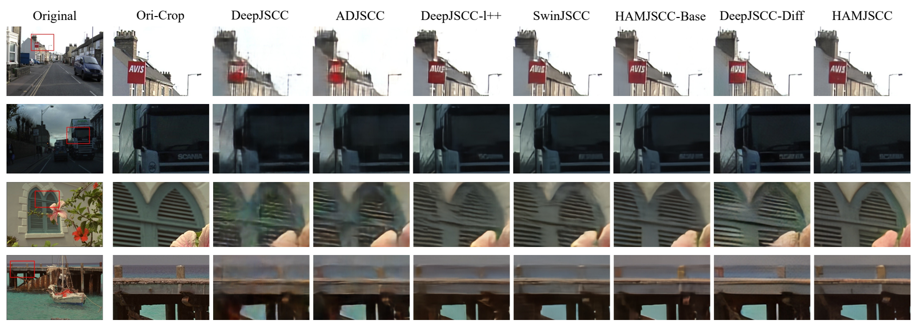

Fig. 10. Comparison of reconstruction results of different schemes on CamVid under AWGN environment and Kodak under Rayleigh environment, with CBR = 1/48. The first and third rows correspond to SNR = 0 dB, and the second and fourth rows correspond to SNR = 3 dB.

As shown in Fig. 9, the left subfigure compares the performance of different schemes under harsh CamVid AWGN environments, where HAMJSCC achieves the lowest LPIPS. The right subfigure presents a global comparison between HAMJSCC and HAMJSCC-Base on Kodak Rayleigh. In the harsh environmental region, although the PSNR decreases slightly, the LPIPS of HAMJSCC is significantly reduced, leading to improved perceptual quality.

As shown in Fig. 10, on the IoV dataset (CamVid), HAMJSCC exhibits stronger sensitivity to key semantic structures in road scenes (e.g., lane lines, curbs, traffic signs), resulting in more complete reconstruction and better perceptual quality. On natural image datasets such as Kodak, HAMJSCC recovers texture details more naturally and clearly, with sharp edges and no visible blockiness or artifacts, demonstrating good cross-scene visual fidelity and generalization capability.

According to Table 2, the decoder latency of HAMJSCC-Base is comparable to that of mainstream Transformer-based schemes (DeepJSCC-l++ and SwinJSCC). Meanwhile, the full model HAMJSCC, through edge-collaborative generative refinement, reduces the decoder latency to 41.06 ms, which is about 75% lower than that of the pure diffusion model DeepJSCC-Diff (165.54 ms).

Table 2. Encoder and decoder complexity comparison

Type	Model	Encoder			Decoder		
		Params (M)	FLOPs (G)	Latency (ms)	Params (M)	FLOPs (G)	Latency (ms)
CNN	DeepJSCC	1.69	0.97	1.02	1.69	3.80	2.19
	ADJSCC	7.20	26.41	18.26	7.20	49.60	25.53
Transformer	DeepJSCC-l++	9.24	16.39	30.30	9.30	17.65	36.85
	SwinJSCC	19.00	17.22	35.01	14.06	16.79	34.75
	HAMJSCC-Base	15.31	17.07	34.58	12.45	16.50	33.74
Generative	DeepJSCC-Diff	1.69	0.97	1.02	559.66	497.91	165.54
	HAMJSCC(Gen)	15.31	17.07	34.58	13.67+3.45	32.93+41.35	12.37+28.69

Note: For HAMJSCC, decoder complexity is edge + device (left: edge, right: device).

Compared with the generative scheme DeepJSCC-Diff, HAMJSCC achieves comparable perceptual quality but higher fidelity and lower inference latency.

5 CONCLUSION

This study proposes HAMJSCC, a hierarchical adaptive multi-path wireless image transmission framework for visual transmission systems. To address complex dynamic environments, it integrates a dual-condition environment-adaptive modulation/demodulation module (SACM/SACD) that jointly exploits SNR and CBR to adjust semantic features. A multi-path decoding mechanism is also designed, incorporating a generative refinement branch under harsh conditions for high-quality edge-device collaborative reconstruction. This framework offers a new solution for IoV visual transmission and a reference for future multi-scenario efficient image transmission.

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